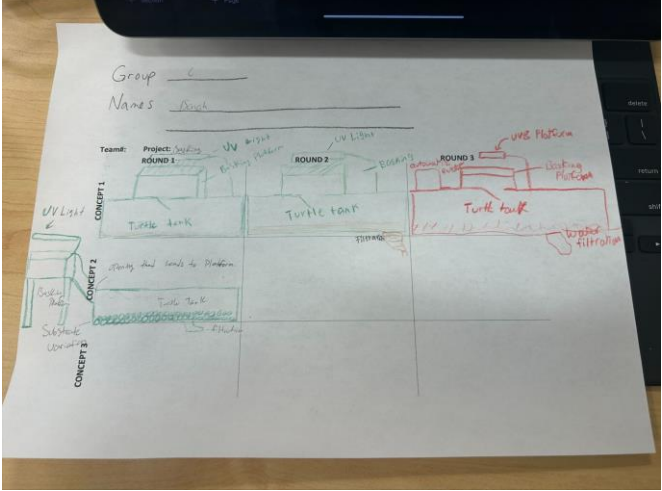
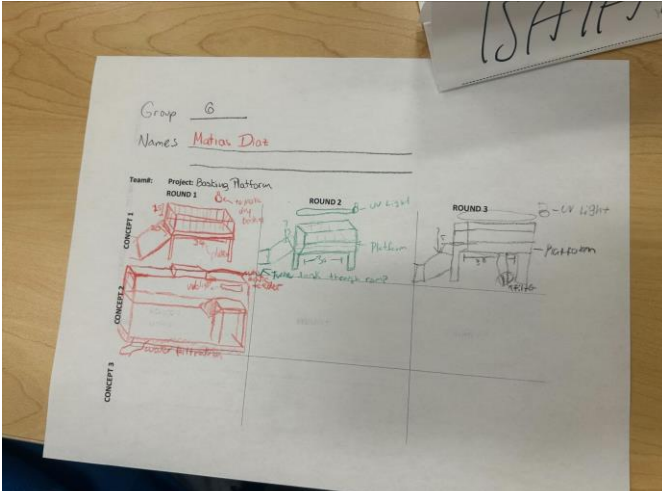


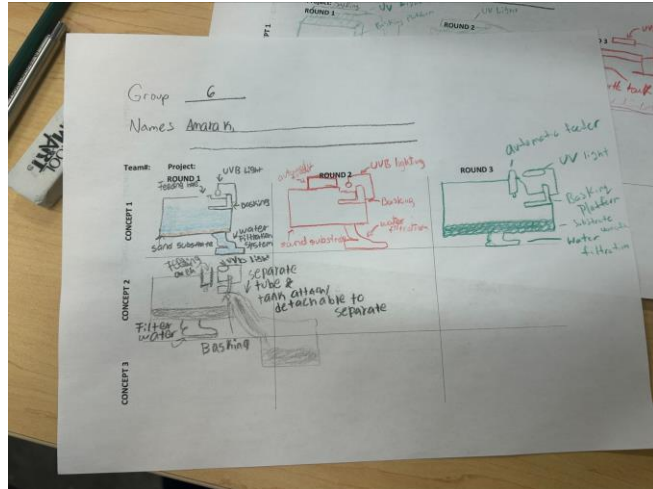
Possible Design Solutions

D1)

Focusing primarily on the basking platform, each of the three of us came up with some ideas on how to modify the design, that also includes other various elements from our other ideas.

Concept	Picture
Multi-level Basking Platform - Similar to the current basking platform, this design is the same concept, but increased dimensions (36x25x15). However, the platform itself is split up into a lower level and higher level so that each turtle has its own space. The lower ramp (6x5) will be big enough so only the male turtle can enter, while the larger ramp (11x7) will allow for the female to enter. This design achieves more room for both turtles to reside in while having their own space.	
Detached Basking Platform - With the dimensions provided in Element C (35x26x15), this design will extend outside of the tank into a separate container which contains the basking platform. The ramps angle would be approximately 30 degrees allowing for the turtles to access the platform more efficiently than they are now.	

Inverted Basking Platform -
The basking platform would be located on the water level of the tank to imitate a turtle coming fresh out of the water to bask on sand. The dimensions would be the same at 36x25x15 to ensure there is sufficient space for both turtles simultaneously.



D2)

Design Matrix (Assisted with co-pilot)

Scale from 1-10 from least viable to most

Design Option	Cost	Ease of Installation	Durability	Accessibility	Maintenance	Average
Multi-level Basking Platform	8	7	8	8	8	7.8
Detached Basking Platform	6	4	7	10	9	7.2
Inverted Basking Platform	5	3	5	3	4	4.0

Our team concluded that design 1 will be the most practical option moving forward, as supported by the rough design matrix. It is positioned in the same spot the current basking platform is so it will not dramatically change the turtles current environment or living patterns and will address the current issue of limited space. With this foundation, there is room for other various aspects that can be added on to improve the design, such as the

multi-level system we implemented. Design 2 although a viable option, is not ideal in terms of installing it. Attia expressed the limited amount of space surrounding the tank so installing it would not be efficient. Lastly, the inverted idea is far-fetched according to our table.

D3)

Plan of Action (assisted with co-pilot)

Task	Duration	Start Date	End Date
Decide what materials to use	1 Day	Nov.6	Nov.6
Elements G,H (Work on the testing plan and priorities)	12 days	Nov.6	No.18
Began building a prototype of our design using cardboard	3 Days	Nov.11	No.13
Element I, begin testing our prototypes and acquiring tangible data to insert into our notebook.	10 Days	Nov.25	Dec.4
Finish prototype, wrap up summaries and reflections	5 Days	Dec.4	Dec.9

Possible Design Materials

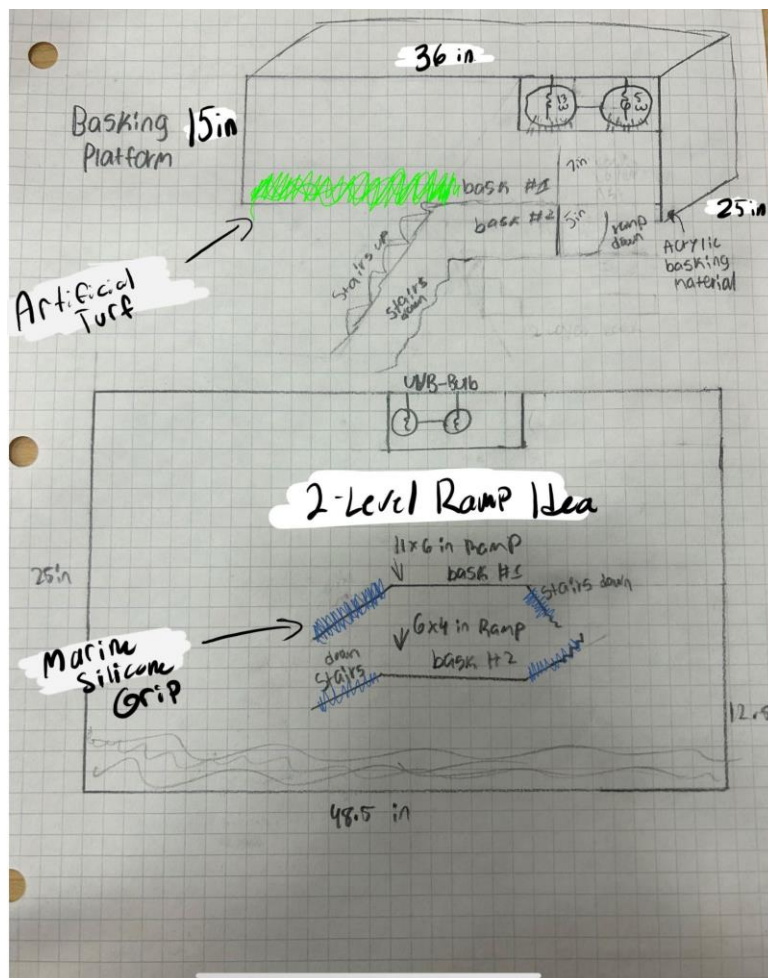
Plastic (lower level walls for structural stability, Upper level floor + upper level ceiling and walls to allow for uv light to pass through)

Rubber mesh (Used for lining both ramps)

D4)

Design Blueprint (Implemented 11/13/25)

These sketches serves as a blueprint for our desired design. The listed dimensions of the basking platform and each of our ramps are provided to give a rough estimate to how our first iteration of design will be.



Dimensions for pieces to laser cut

(Material: Acrylic Plastic sheets, all measurements in inches)

Basking platform (36x25x15")

Panels

- (25x15")

- Count: 2

- (36x25")

- Count: 3

- (18x25")

- Count: 1

- Ramp 1 (12x7")

- Count: 2

- Ramp 2 (7x5")

Count 2

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